

DATA STRUCTURES: MASTER TEXTBOOK

1. Introduction & Complexity

Algorithm Analysis

Data structures are ways to organize data for efficient access. Complexity is measured using **Big O notation** indicating time and space performance as input size grows (e.g., $O(1)$, $O(n)$, $O(\log n)$).

2. Linear Data Structures

Arrays, Lists, Stacks, Queues

- **Arrays:** Fixed-size, contiguous memory. Fast indexing.
- **Linked Lists:** Dynamic nodes with pointers. Efficient insertions/deletions.
- **Stack (LIFO):** Last-In-First-Out structure (e.g., function calls).
- **Queue (FIFO):** First-In-First-Out structure (e.g., print buffers).

3. Non-Linear Data Structures

Trees and Graphs

Trees: Hierarchical structure (e.g., Binary Search Trees). Ideal for searching and sorting.

Graphs: Networks of nodes (vertices) and edges. Used to model connections (e.g., Social Networks, Maps).

4. Searching and Sorting

Efficiency Matters

Basic algorithms include **Binary Search** ($O(\log n)$) for sorted data, and sorting algorithms like **QuickSort** and **MergeSort** which optimize to $O(n \log n)$.